

Total No. of Questions : 8]

SEAT No. :

P6749

[6181] - 337

[Total No. of Pages : 2

B.E. (Mechanical/ Automobile) (Honors)
MODELLING AND SIMULATION OF EHV
(2019 Pattern) (Semester - VII) (402034MJ)

Time : 2 ½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.No.1 or Q.No.2, Q.No.3 or Q.No.4, Q.No.5 or Q.No.6, Q.No.7 or Q.No.8.
- 2) Figures to the right indicate full marks.
- 3) Draw the neat sketch wherever necessary.

Q1) a) Explain A Motor Control Unit in electric vehicle with its functional features? **[4]**

b) Explain the following concepts related to modes of BMS control, **[8]**

- i) Centralized BMS Architecture
- ii) Modular BMS Topology
- iii) Primary/Subordinate BMS
- iv) Distributed BMS Architecture

c) What is Battery/Cell Control System in EV? Explain with its functions and working principal? **[8]**

OR

Q2) a) Explain Torque and speed coupling in electric vehicle? **[4]**

b) Explain following motor Control methods (any three) **[8]**

- i) Phase flux linkage
- ii) Phase Inductance
- iii) Modulated Signal Injection
- iv) Mutually Induced Voltage
- v) Observer based Theory

c) Explain the importance of Sensor Management and Integration for electric vehicle with functional and application point of view? **[8]**

P.T.O.

- Q3) a)** Explain, “ Placement of Motors” in electric vehicle system? [8]
b) Explain the important feature of following (any four), [8]
i) Unicycle
ii) Bicycle
iii) Dicycle
iv) Tricycle
v) Quadracycle

OR

- Q4) a)** Explain, “Battery and Motion Transmission Systems” in electric vehicle system? [8]
b) Explain Propulsion and Power distribution system in electric vehicle with its main components? [8]

- Q5) a)** Explain Design and repair tips for frame of vehicles? [8]
b) Explain the types and components of battery pack in electric vehicle? [8]

OR

- Q6) a)** Explain the types of chassis on basis of following. [8]
i) Engine Location
ii) Assembly
iii) Frame
b) Explain the following parameters related to vehicle dynamics in detail. [8]
i) Aerodynamic drag
ii) Aerodynamic lift
iii) Side force
iv) Rolling movement

- Q7) a)** List out the feature of Electric Vehicle Dashboard considering suitable example of 2W/3W/4W. [8]
b) What do you mean by ergonomics and aesthetics aspects of ev? Explain in detail with important features? [10]

OR

- Q8) a)** Explain Noise Factors & Failure Modes in electric vehicle? [10]
b) Explain the important features of FMEA analysis in electric vehicle? [8]

